CCR-k General Theory: Higher-Order Chronological Carry, Hessenberg Hierarchies, and Universal Stirling Shadows

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Abstract

We develop the CCR-k General Theory, an all-order chronological-carry hierarchy motivated by an ordered correction process arising in deterministic pushdown normalization. The geometric input is deliberately separated from the arithmetic one: the correction chronology supplies a path, hence exactly one connected interval support type at every order, while the numerical weight of that support is additional typed arithmetic data and is not claimed to be forced uniquely by the underlying automaton. Starting from the third-order chronological carry recurrence, we define a terminal-block CCR-k class with one source sequence at each new order. We prove all-order terminal-block separation, an exact higher-band lower-Hessenberg determinant realization, and a subdiagonal light-cone theorem showing that the r-th fixed column sees no band deeper than the $(r-1)$-st subdiagonal. Falling-factorial normalization yields first-order transport equations and an inverse formula reconstructing every connected source from adjacent normalized-column rows. In the stationary p-divisible subclass, the extension space through order k is the integer parameter lattice $\mathbb{Z}^{k-3}$; the primitive same-sign normalization is the point with every higher parameter equal to one. The associated operator is

$$\mathcal{H}^{[k]}_{p,\boldsymbol{\lambda}} = D + L_p - pXL_p - p\sum_{r=3}^{k} \lambda_r X^{r-1}.$$

We further prove arbitrary-order polynomiality of the normalized fixed columns and a universal raw mod-p shadow

$$P^{[k]}_{p,n}(u) \equiv \prod_{j=1}^{n}(1+ju) \pmod{(p, u^{k+1})},$$

so all p-divisible connected parameters disappear in the unsigned first-kind Stirling shadow even though the exact columns identify them. The third- and fourth-order theories are recovered exactly as special cases, including the independent fourth-order parameter and its constant normalized-column translation.

Keywords. chronological carry refinement; deterministic pushdown automata; connected interval blocks; lower-Hessenberg matrices; production matrices; fixed-column transport; Stirling numbers of the first kind; identifiability.

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1 Introduction

The purpose of this article is to formulate and prove the *CCR-k General Theory*. Chronological carry refinement began as an arithmetic lift of a concrete correction chronology: a deterministic pushdown normalization process singles out correction events, and those events inherit a linear time order. Once the events are placed on a path, connected supports become rigid. A connected set of r chronological positions is necessarily a consecutive interval of length r. What does *not* follow from this geometry is a unique numerical weight for that interval. The third-order arithmetic lift and the fourth-order extension problem were previously studied separately [1, 2], within the broader normalization program developed in the research monograph [3]. The present article is nevertheless fully self-contained: no definition, theorem, or proof below is imported from those works without being stated or proved here.

This distinction is essential. At order three, a particular arithmetic polarization leads to the recurrence

$$\begin{aligned} P^{[3]}_{p,n}(u) &= (1+\ell_n u)P^{[3]}_{p,n-1}(u) + p(n-1)\ell_{n-1}u^2 P^{[3]}_{p,n-2}(u) \\ &\quad - p(n-2)(n-1)u^3 P^{[3]}_{p,n-3}(u) \pmod{u^4}, \end{aligned} \tag{1}$$

where

$$\ell_n = (p+1)n - p. \tag{2}$$

At order four, however, the complete third jet and path locality leave one genuinely new arithmetic coordinate. In the stationary p-divisible class the extension is

$$\begin{aligned} P^{[4,\lambda]}_{p,n}(u) &= (1+\ell_n u)P^{[4,\lambda]}_{p,n-1}(u) + p(n-1)\ell_{n-1}u^2 P^{[4,\lambda]}_{p,n-2}(u) \\ &\quad - p(n-2)(n-1)u^3 P^{[4,\lambda]}_{p,n-3}(u) \\ &\quad + \lambda p(n-3)(n-2)(n-1)u^4 P^{[4,\lambda]}_{p,n-4}(u) \pmod{u^5}. \end{aligned} \tag{3}$$

Thus chronology fixes the *support type*, while arithmetic polarization supplies the new coefficient.

The purpose of this paper is to put that phenomenon into an all-order theorem. We work with an explicit terminal-block class in which a connected terminal interval of size r contributes a typed source at degree r. Within this class we prove the following chain:

$$\boxed{\begin{array}{c} \text{one connected interval type at order } r \\ \Updownarrow \\ \text{one new typed recurrence source at order } r \\ \Updownarrow \\ \text{one new lower-Hessenberg subdiagonal at order } r \\ \Updownarrow \\ \text{one new source term in normalized-column transport.} \end{array}}$$

The exact source is visible integrally and is reconstructible from the normalized column, but every p-divisible source is invisible in the universal mod-p shadow.

Main contributions

The paper proves the following results.

1. **All-order terminal-block separation.** Once all sources below order r are fixed, the only new degree-r datum is the connected terminal r-block source.
2. **Exact higher-band lower-Hessenberg realization.** The order-r source is placed on the $(r-1)$-st subdiagonal with an explicit rising-factorial coefficient, and determinant expansion reproduces the recurrence exactly.
3. **Column light cone.** In a lower-Hessenberg determinant, the coefficient of u^j cannot see subdiagonals deeper than $j-1$.
4. **Transport and inverse transport.** Falling-factorial normalization converts coefficient extraction into a first-order row transport system; the order-r source is reconstructed exactly from adjacent rows once lower sources are known.
5. **Stationary p-divisible parameterization.** Through order k, the stated stationary subclass is coordinatized by $\mathbb{Z}^{k-3}$, with one integer parameter λ_r for every $r = 4, \ldots, k$. A primitive same-sign convention selects $\lambda_r = 1$ for all r. The substantive rigidity comes from the exact transport, reconstruction, and visibility results that make these coordinates observable.
6. **Operator realization.** In the divided-power basis, the full stationary hierarchy is represented by a finite truncation of
$$D + L_p - pXL_p - p\sum_{r=3}^{k} \lambda_r X^{r-1}.$$
7. **Polynomial fixed columns.** Every stationary normalized fixed column is a polynomial of exact degree equal to the column index, with leading coefficient $(p+1)^j/(2^j j!)$.
8. **Universal Stirling shadow.** All p-divisible interaction data vanish modulo p, leaving the free product $\prod_{j=1}^{n}(1+ju)$. Thus the raw fixed-column shadow is a near-diagonal unsigned Stirling number of the first kind.

The scope is deliberately narrow. We do not claim a new general theory of cumulants, carries, Stirling numbers, or Hessenberg matrices. Those are classical neighboring structures; the contribution is the exact all-order classification and identifiability package for the chronological terminal-block family defined below.

2 Chronological support geometry and scope

Fix a prime $p \geq 5$ unless stated otherwise. The structural determinant and transport identities are polynomial identities and therefore make sense more generally, but the prime hypothesis is natural for the mod-p arithmetic statements.

The chronology is represented by the path graph on consecutive correction positions. The following elementary lemma is the geometric reason that only one connected support *type* appears at each order.

Lemma 2.1 (Connected interval support)**.** *Every connected subset of a path having cardinality r is a consecutive interval*
$$\{j, j+1, \ldots, j+r-1\}.$$
Hence there is exactly one connected support type of size r.

Proof. If $a < c$ lie in a connected subset S, then the unique path from a to c contains every integer between them. Connectivity forces each of those intermediate vertices into S. Thus S is an interval, and its cardinality determines its length. $\square$

Remark 2.2 (Geometry does not determine arithmetic)**.** *Lemma 2.1 determines the support of a new connected interaction, not its numerical weight. The fourth-order family* (3) *already shows that a complete lower jet plus chronological locality need not identify the next coefficient. Throughout this paper, numerical connected weights are therefore treated as explicitly typed arithmetic data.*

Figure 1 summarizes the structural dictionary used throughout the paper.

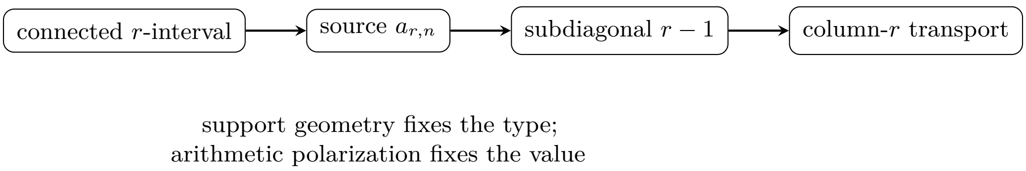

Figure 1: The order/band/transport dictionary. The first arrow is a modeling interface: path geometry supplies one connected support type, while the numerical source remains independent arithmetic data.

3 The third- and fourth-order prototypes

We record the low-order structures that motivate the general definitions. Set $P_{p,0} = 1$ and $P_{p,n} = 0$ for $n < 0$.

3.1 CCR-3

The third-order recurrence is (1). It has the lower-Hessenberg realization with

$$(H_p)_{i,i} = \ell_i, \qquad (H_p)_{i,i+1} = 1, \tag{4}$$

$$(H_p)_{i+1,i} = -pi\ell_i, \qquad (H_p)_{i+2,i} = -pi(i+1), \tag{5}$$

and no deeper subdiagonal. If $H_{p,n}$ denotes the $n \times n$ leading principal truncation, then

$$\det(I_n + uH_{p,n})$$

satisfies (1) exactly.

In the divided-power basis

$$e_i(x) = \frac{x^{i-1}}{(i-1)!}, \qquad i \geq 1,$$

put

$$D = \partial_x, \qquad Xf = xf, \qquad N = XD, \qquad L_p = I + (p+1)N.$$

Then

$$De_i = e_{i-1}, \qquad Xe_i = ie_{i+1}, \qquad L_pe_i = \ell_ie_i,$$

and the matrix (4)–(5) is the matrix of

$$\mathcal{H}_p = D + L_p - pXL_p - pX^2. \tag{6}$$

3.2 Order four and the first independent higher coordinate

The recurrence (3) adds exactly one new connected source type, supported on a consecutive quadruple. The numerical coefficient λ is independent data.

Proposition 3.1 (Fourth-order non-identifiability). *For every integer sequence $(q_{p,n})_{n\geq 4}$, the recurrence*

$$\begin{aligned} P_{p,n}(u) &= (1+\ell_n u)P_{p,n-1}(u) + p(n-1)\ell_{n-1}u^2 P_{p,n-2}(u) \\ &\quad - p(n-2)(n-1)u^3 P_{p,n-3}(u) + q_{p,n}u^4 P_{p,n-4}(u) \pmod{u^5} \end{aligned}$$

has the same coefficients through degree three as CCR-3. Hence CCR-3 plus chronological locality does not determine a unique fourth-order numerical weight.

Proof. The new term is divisible by u^4, so it cannot affect coefficients of degree at most three. Induction on n gives equality of the complete third jet for every choice of $q_{p,n}$. □

For the stationary p-divisible choice

$$q_{p,n} = \lambda p n - 1_{\underline{3}},$$

the new Hessenberg band is

$$(H_{p,\lambda})_{i+3,i} = -\lambda p\, i(i+1)(i+2),$$

and the operator becomes

$$\mathcal{H}_{p,\lambda} = D + L_p - pXL_p - pX^2 - \lambda p X^3. \tag{7}$$

This is the prototype for the arbitrary-order operator in section 9.

4 General terminal-block CCR-k

Fix $k \geq 3$ and work in the truncated ring

$$R_k = \mathbb{Z}[u]/(u^{k+1}).$$

For $r \geq 3$, let $a_{r,n} \in \mathbb{Z}$ be the normalized connected source at order r, after removal of the chronological placement factor

$$n - 1_{\underline{r-1}} = (n-1)(n-2)\cdots(n-r+1).$$

The third-order baseline fixes

$$a_{3,n} = -p. \tag{8}$$

Definition 4.1 (Typed terminal-block CCR-k). *A typed terminal-block* CCR-k *family is a sequence $P^{[k]}_{p,n}(u) \in R_k$ satisfying*

$$\boxed{\begin{aligned} P^{[k]}_{p,n}(u) &= (1+\ell_n u)P^{[k]}_{p,n-1}(u) \\ &\quad + p(n-1)\ell_{n-1}u^2 P^{[k]}_{p,n-2}(u) \\ &\quad + \sum_{r=3}^{k} n - 1_{\underline{r-1}}\, a_{r,n} u^r P^{[k]}_{p,n-r}(u) \pmod{u^{k+1}}, \end{aligned}} \tag{9}$$

with $P_{p,0} = 1$, $P_{p,n} = 0$ for $n < 0$, and (8).

The definition is intentionally explicit: it is the class of chronological recurrences that factor by the size of the terminal connected interval block. It does not assert that every conceivable path-local model must have this factorization.

Write

$$C_{n,j} = [u^j]P^{[k]}_{p,n}(u).$$

Theorem 4.2 (All-order terminal-block separation)**.** *For every $r \le k$,*

$$\boxed{\begin{aligned} C_{n,r} &= C_{n-1,r} + \ell_n C_{n-1,r-1} + p(n-1)\ell_{n-1}C_{n-2,r-2} \\ &\quad + \sum_{s=3}^{r-1} n-1_{\underline{s-1}}\, a_{s,n}C_{n-s,r-s} + n-1_{\underline{r-1}}\, a_{r,n}. \end{aligned}} \tag{10}$$

After all sources of orders below r have been fixed, the only new recurrence datum entering degree r is $a_{r,n}$.

Proof. Extract u^r from (9). A terminal source of size s contributes

$$n-1_{\underline{s-1}}\, a_{s,n}[u^{r-s}]P^{[k]}_{p,n-s}(u).$$

When $s < r$, this depends only on a coefficient of degree $r - s < r$, hence only on lower-order data. For the $s = r$ term there are two boundary regimes. If $1 \le n < r$, then the placement factor $n-1_{\underline{r-1}}$ contains a zero factor, so the entire term vanishes. If $n \ge r$, then $P_{p,n-r}$ has constant term one (inductively from $P_{p,0} = 1$ and the recurrence), so $[u^0]P_{p,n-r} = 1$ and the final term in (10) follows. Sources of order greater than r are divisible by a power of u too large to contribute. $\square$

Corollary 4.3 (All-order non-identifiability)**.** *Two typed terminal-block families that agree through order $r-1$ but differ in $a_{r,n}$ have identical $(r-1)$-jets. Thus lower-order data do not identify the next connected arithmetic source.*

Proof. The first term involving $a_{r,n}$ has degree r, so all lower coefficients are independent of it. $\square$

5 Higher-band lower-Hessenberg realization

For $i \ge 1$ and $m \ge 0$, write

$$i^{\overline{m}} = i(i+1)\cdots(i+m-1), \qquad i^{\overline{0}} = 1.$$

Definition 5.1 (General Hessenberg matrix)**.** *Let $H^{[k]}$ be the infinite lower-Hessenberg matrix with*

$$H_{i,i} = \ell_i, \qquad H_{i,i+1} = 1, \tag{11}$$

$$H_{i+1,i} = -pi\ell_i, \tag{12}$$

and, for $3 \le r \le k$,

$$\boxed{H_{i+r-1,i} = (-1)^{r-1} i^{\overline{r-1}}\, a_{r,i+r-1}.} \tag{13}$$

All entries below the $(k-1)$-st subdiagonal are zero. Let $H^{[k]}_n$ denote the leading $n \times n$ principal truncation.

Theorem 5.2 (Exact Hessenberg realization)**.** *The determinant polynomials*

$$D_n^{[k]}(u) = \det(I_n + uH_n^{[k]})$$

satisfy (9) *exactly. Therefore*

$$\boxed{D_n^{[k]}(u) = P_{p,n}^{[k]}(u) \quad \textit{in } R_k.}$$

Proof. Expand $\det(I_n + uH_n^{[k]})$ along the last row. The diagonal term contributes

$$(1 + \ell_n u)D_{n-1}^{[k]}.$$

The first subdiagonal entry is $-p(n-1)\ell_{n-1}$. Its cofactor sign is negative, and the remaining forced superdiagonal closure contributes one factor of u, so this term is

$$+p(n-1)\ell_{n-1}u^2 D_{n-2}^{[k]}.$$

Now choose the entry in column $n - r + 1$, where $r \geq 3$. By (13),

$$H_{n,n-r+1} = (-1)^{r-1} n - r + 1^{\overline{r-1}} a_{r,n}.$$

The cofactor sign is $(-1)^{r-1}$. Lower-Hessenberg structure forces the last $r-1$ columns of the cofactor to close through the consecutive unit superdiagonal entries, contributing u^{r-1}; the selected last-row entry contributes the remaining factor u. The two signs cancel, and

$$n - r + 1^{\overline{r-1}} = n - 1_{\underline{r-1}}.$$

The untouched leading block is $I_{n-r} + uH_{n-r}^{[k]}$. Hence the contribution is

$$n - 1_{\underline{r-1}} a_{r,n} u^r D_{n-r}^{[k]}.$$

Summing the available bands gives (9). The initial conditions agree, so the two families coincide. □

The theorem turns the chronological order filtration into an exact band filtration. The next result shows that this filtration is also visible coefficientwise.

6 The subdiagonal light cone

Theorem 6.1 (Subdiagonal light cone)**.** *Let $M = (m_{ij})$ be any lower-Hessenberg matrix, so $m_{ij} = 0$ for $j > i + 1$, and let*

$$Q_n(u) = \det(I_n + uM_n).$$

Then $[u^j]Q_n(u)$ depends only on the diagonal, superdiagonal, and first $j-1$ subdiagonals. Equivalently, an entry on the s-th subdiagonal can first affect degree $s + 1$.

Proof. The coefficient $[u^j]Q_n(u)$ is the sum of the principal $j \times j$ minors of M_n. Suppose a permutation term in such a minor uses an entry on the s-th subdiagonal. Relative to the diagonal, that move has displacement $-s$. Because M is lower-Hessenberg, every positive displacement in the same permutation is at most $+1$. Since the total permutation displacement is zero, compensating $-s$ requires at least s positive unit displacements. The permutation therefore moves at least $s+1$ indices and cannot occur in a minor smaller than $(s+1)\times(s+1)$. Thus the s-th subdiagonal is invisible below degree $s + 1$. □

Corollary 6.2 (Exact order filtration)**.** *In the typed CCR hierarchy, the order-r source affects no fixed column below r, while every band deeper than $r-1$ is invisible to the r-th fixed column.*

This provides a matrix-theoretic proof that higher interactions cannot retroactively change lower jets.

7 Normalized fixed-column transport and inverse reconstruction

For $j \geq 0$, set
$$C_{n,j} = [u^j]P^{[k]}_{p,n}(u).$$
For $m \geq j+1$, define the falling-factorial normalized column
$$\boxed{\Pi_{p,j}(m) = \frac{C_{m-1,j}}{m-1_{\underline{j}}}.} \tag{14}$$
Also set $\Pi_{p,0}(m) = 1$.

The first admissible row $m = j$ must be treated as a boundary equation; this avoids the meaningless expression $0 \cdot \Pi_{p,j}(j)$.

Theorem 7.1 (Boundary and transport equations)**.** *For $1 \leq j \leq k$, the first normalized row is determined by*
$$\boxed{\begin{aligned} j\Pi_{p,j}(j+1) = \ell_j\Pi_{p,j-1}(j) + p\ell_{j-1}\Pi_{p,j-2}(j-1)& \\ + \sum_{r=3}^{j} a_{r,j}\Pi_{p,j-r}(j-r+1),& \end{aligned}} \tag{15}$$
where terms with negative column index are omitted. For every $m \geq j+1$,
$$\boxed{\begin{aligned} m\Pi_{p,j}(m+1) = (m-j)\Pi_{p,j}(m) + \ell_m\Pi_{p,j-1}(m)& \\ + p\ell_{m-1}\Pi_{p,j-2}(m-1)& \\ + \sum_{r=3}^{j} a_{r,m}\Pi_{p,j-r}(m-r+1).& \end{aligned}} \tag{16}$$

Proof. Extract u^j from (9) with $n = m$:
$$\begin{aligned} C_{m,j} = C_{m-1,j} &+ \ell_m C_{m-1,j-1} + p(m-1)\ell_{m-1}C_{m-2,j-2} \\ &+ \sum_{r=3}^{j} m-1_{\underline{r-1}}a_{r,m}C_{m-r,j-r}. \end{aligned}$$
Use
$$\begin{aligned} C_{m,j} &= m_{\underline{j}}\Pi_{p,j}(m+1) = mm-1_{\underline{j-1}}\Pi_{p,j}(m+1), \\ C_{m-1,j} &= m-1_{\underline{j}}\Pi_{p,j}(m) = (m-j)m-1_{\underline{j-1}}\Pi_{p,j}(m), \end{aligned}$$
and
$$m-1_{\underline{r-1}}m-r_{\underline{j-r}} = m-1_{\underline{j-1}}.$$
For $m \geq j+1$, division by $m-1_{\underline{j-1}}$ gives (16). At $m = j$, the term $C_{j-1,j}$ is zero because its degree exceeds its row size, so the same coefficient extraction yields (15) without ever defining $\Pi_{p,j}(j)$. □

The transport equation is triangular in the interaction order and therefore invertible.

Theorem 7.2 (Exact source reconstruction)**.** *Fix $r \geq 3$ and suppose all sources of orders below r are known. The first source value is recovered from the boundary row by*

$$\boxed{\begin{aligned} a_{r,r} &= r\Pi_{p,r}(r+1) - \ell_r \Pi_{p,r-1}(r) - p\ell_{r-1}\Pi_{p,r-2}(r-1) \\ &\quad - \sum_{s=3}^{r-1} a_{s,r}\Pi_{p,r-s}(r-s+1). \end{aligned}} \tag{17}$$

For every $m \geq r+1$,

$$\boxed{\begin{aligned} a_{r,m} &= m\Pi_{p,r}(m+1) - (m-r)\Pi_{p,r}(m) - \ell_m \Pi_{p,r-1}(m) \\ &\quad - p\ell_{m-1}\Pi_{p,r-2}(m-1) \\ &\quad - \sum_{s=3}^{r-1} a_{s,m}\Pi_{p,r-s}(m-s+1). \end{aligned}} \tag{18}$$

Thus the complete normalized source sequence is identifiable from the r-th column together with the already known lower-order data.

Proof. Set $j = r$ in theorem 7.1. The term with source order r is multiplied by $\Pi_{p,0} = 1$. Solving the boundary and transport equations for that source gives (17) and (18). □

8 Stationary p-divisible extensions

The preceding theory allows arbitrary row-dependent source sequences. A particularly rigid and useful class is obtained by removing that row dependence after the chronological placement factor.

Definition 8.1 (Stationary p-divisible class)**.** *For $r \geq 4$, a source is stationary p-divisible if*

$$a_{r,n} = (-1)^r p\lambda_r$$

for an integer λ_r independent of n. We set $\lambda_3 = 1$, so that $a_{3,n} = -p$ agrees with CCR-3.

Theorem 8.2 (Stationary extension classification)**.** *A stationary p-divisible extension through order k is uniquely determined by the parameter vector*

$$\boxed{(\lambda_4, \lambda_5, \ldots, \lambda_k) \in \mathbb{Z}^{k-3}.}$$

Its recurrence is

$$\boxed{\begin{aligned} P_{p,n}^{[k;\boldsymbol{\lambda}]}(u) &= (1+\ell_n u)P_{p,n-1} + p(n-1)\ell_{n-1}u^2 P_{p,n-2} \\ &\qquad + p\sum_{r=3}^{k}(-1)^r \lambda_r n - 1_{\underline{r-1}} u^r P_{p,n-r} \pmod{u^{k+1}}, \end{aligned}} \tag{19}$$

with $\lambda_3 = 1$.

Proof. By Lemma 2.1, there is one connected support type at each order. By theorem 4.2, a typed terminal-block model introduces one new normalized source sequence at that order. Stationarity makes the normalized source independent of n, and p-divisibility expresses it uniquely as p times an integer. The factor $(-1)^r$ is a sign convention chosen so that positive λ_r produces a common negative sign on the corresponding Hessenberg subdiagonal; see equation (21) below. □

Definition 8.3 (Primitive same-sign hierarchy)**.** *The primitive same-sign normalization is the stationary point*

$$\boxed{\lambda_r = 1 \qquad (3 \le r \le k).}$$

Thus

$$\boxed{\begin{aligned} P_{p,n}^{[k],\mathrm{prim}}(u) &= (1+\ell_n u)P_{p,n-1} + p(n-1)\ell_{n-1}u^2 P_{p,n-2} \\ &\quad + p\sum_{r=3}^{k}(-1)^r n-1_{\underline{r-1}}u^r P_{p,n-r} \pmod{u^{k+1}}. \end{aligned}} \tag{20}$$

The word *primitive* here names a selected arithmetic normalization. It is not a claim that the underlying pushdown transition table forces $\lambda_r = 1$.

9 Operator realization

Under Definition 8.1, the deeper Hessenberg bands simplify to

$$\boxed{H_{i+r-1,i} = -\lambda_r p\, i^{\overline{r-1}}, \qquad 3 \le r \le k.} \tag{21}$$

All higher connected bands therefore have the same matrix sign when $\lambda_r > 0$.

In the divided-power basis $e_i = x^{i-1}/(i-1)!$, one has

$$X^s e_i = i^{\overline{s}} e_{i+s}.$$

Theorem 9.1 (General differential-operator realization)**.** *The stationary* CCR-k *Hessenberg matrix is the matrix of*

$$\boxed{\mathcal{H}_{p,\boldsymbol{\lambda}}^{[k]} = D + L_p - pXL_p - p\sum_{r=3}^{k}\lambda_r X^{r-1}.} \tag{22}$$

For the primitive same-sign hierarchy,

$$\boxed{\mathcal{H}_{p,\mathrm{prim}}^{[k]} = D + L_p - pXL_p - p(X^2 + X^3 + \cdots + X^{k-1}).} \tag{23}$$

Proof. The terms D, L_p, and $-pXL_p$ reproduce the unit superdiagonal, diagonal ℓ_i, and first lower band $-pi\ell_i$. For $r \ge 3$,

$$-p\lambda_r X^{r-1}e_i = -p\lambda_r i^{\overline{r-1}} e_{i+r-1},$$

which is exactly (21). □

10 Polynomial fixed columns

The normalized transport system has a useful consequence not visible from the determinant alone: stationary fixed columns are polynomial at every order.

For $j \ge 1$, define the linear operator on $\mathbb{Q}[m]$

$$(\mathcal{T}_j f)(m) = mf(m+1) - (m-j)f(m). \tag{24}$$

If $f(m) = c_d m^d + \cdots$, then

$$\mathcal{T}_j f = (d+j)c_d m^d + \text{lower-degree terms}. \tag{25}$$

Thus $\mathcal{T}_j$ preserves degree and is an automorphism of $\mathbb{Q}[m]$.

Theorem 10.1 (Arbitrary-order fixed-column polynomiality)**.** *For every stationary p-divisible* CCR-k *family and every $0 \le j \le k$, the normalized column $\Pi_{p,j}(m)$ is the restriction, for $m \ge j+1$, of a unique polynomial in*

$$\mathbb{Q}[p, \lambda_4, \ldots, \lambda_j][m].$$

For $j \ge 1$, its degree is exactly j, and its leading coefficient is

$$\boxed{[m^j]\Pi_{p,j}(m) = \frac{(p+1)^j}{2^j j!}.} \tag{26}$$

Proof. Proceed by induction on j. The case $j = 0$ is $\Pi_{p,0} = 1$. Suppose the assertion holds below j. Under stationarity, the right-hand side of (16) excluding the term $(m-j)\Pi_{p,j}(m)$ is a polynomial $G_j(m)$. Its only degree-j contribution comes from

$$\ell_m \Pi_{p,j-1}(m),$$

because the pair term has degree at most $j-1$, while the source term of order r multiplies a column of degree $j-r$. Hence G_j has degree j.

Equation (16) is precisely

$$\mathcal{T}_j \Pi_{p,j} = G_j.$$

By (25), $\mathcal{T}_j$ is triangular with nonzero diagonal $d+j$ on the monomial basis, so it is invertible on $\mathbb{Q}[m]$. There is therefore a unique polynomial solution of degree j. At $m = j$, the polynomial identity reduces to the boundary equation (15), so the polynomial solution has the same first admissible value as the normalized column. The first-order recurrence then identifies the two for all $m \ge j+1$.

If c_{j-1} is the leading coefficient of $\Pi_{p,j-1}$, the leading coefficient of G_j is $(p+1)c_{j-1}$. Since $\mathcal{T}_j$ multiplies the leading coefficient of a degree-j polynomial by $2j$,

$$c_j = \frac{p+1}{2j} c_{j-1}.$$

Starting from $c_0 = 1$ gives (26). ☐

The leading term is independent of every higher connected parameter. Those parameters change lower-degree structure while leaving the universal top-degree growth fixed.

11 Exact parameter translation and identifiability

Suppose two stationary families agree at every order below r but have order-r parameters λ_r and μ_r. Let

$$\Delta_m = \Pi_{p,r}^{(\lambda_r)}(m) - \Pi_{p,r}^{(\mu_r)}(m).$$

Theorem 11.1 (Stationary translation law)**.** *The first admissible row satisfies*

$$\boxed{\Delta_{r+1} = \frac{(-1)^r p(\lambda_r - \mu_r)}{r}.} \tag{27}$$

For $m \ge r+1$,

$$\boxed{m\Delta_{m+1} = (m-r)\Delta_m + (-1)^r p(\lambda_r - \mu_r).} \tag{28}$$

The unique solution is the constant translation

$$\boxed{\Delta_m = \frac{(-1)^r p(\lambda_r - \mu_r)}{r} \qquad (m \ge r+1).} \tag{29}$$

Proof. Subtract the two order-r instances of theorem 7.1. Every lower-order term cancels, leaving only the difference

$$(-1)^r p(\lambda_r - \mu_r).$$

The boundary equation gives (27); the ordinary transport rows give (28). The constant in (29) satisfies both equations, and the first-order recurrence has a unique solution after its first value is fixed. □

Corollary 11.2 (One-row identifiability)**.** *Once lower parameters are known, a single exact normalized observation in column r identifies the stationary parameter λ_r.*

At $r = 4$, equation (29) becomes the recovered fourth-order translation

$$\Pi_{p,4}^{(\lambda)}(m) - \Pi_{p,4}^{(\mu)}(m) = \frac{p(\lambda - \mu)}{4}.$$

At $r = 5$, it predicts the alternating next case

$$\Pi_{p,5}^{(\lambda_5)}(m) - \Pi_{p,5}^{(\mu_5)}(m) = -\frac{p(\lambda_5 - \mu_5)}{5}.$$

12 Universal Stirling shadows

Assume now that every connected source is divisible by p; in particular this holds throughout the stationary class. Since

$$\ell_n = n + p(n-1) \equiv n \pmod{p},$$

and every interaction term beyond the singleton term has an explicit factor of p, the full recurrence collapses modulo p.

Theorem 12.1 (Arbitrary-order raw Stirling shadow)**.** *For every $k \geq 3$,*

$$\boxed{P_{p,n}^{[k]}(u) \equiv \prod_{j=1}^{n}(1 + ju) \pmod{(p, u^{k+1})}.} \tag{30}$$

Consequently, for $0 \leq r \leq k$,

$$\boxed{[u^r]P_{p,n}^{[k]}(u) \equiv e_r(1, 2, \ldots, n) = \begin{bmatrix} n+1 \\ n+1-r \end{bmatrix} \pmod{p},} \tag{31}$$

where the bracket denotes an unsigned Stirling number of the first kind.

Proof. Modulo p, recurrence (9) becomes

$$P_{p,n}^{[k]}(u) \equiv (1 + nu)P_{p,n-1}^{[k]}(u) \pmod{(p, u^{k+1})}.$$

Iterating from $P_{p,0} = 1$ gives (30). Coefficient extraction gives the elementary symmetric function in $1, \ldots, n$, which is the displayed near-diagonal unsigned Stirling number. □

The raw form (31) is the safest universal theorem because it requires no modular division. The normalized shadow admits a precise rational-polynomial formulation.

Proposition 12.2 (Normalized shadow and denominator resonances)**.** *For fixed r, define*

$$S_r(m) = \frac{e_r(1, 2, \ldots, m-1)}{m - 1_{\underline{r}}}. \tag{32}$$

Then $S_r(m) \in \mathbb{Q}[m]$ has degree r. In the stationary p-divisible hierarchy,

$$\Pi_{p,r}(m) - S_r(m) \in p\,\mathbb{Q}[p, \lambda_4, \ldots, \lambda_r, m]. \tag{33}$$

Hence there exists an integer $D_r > 0$, depending only on r, such that for every prime $p \nmid D_r$ the normalized polynomial congruence

$$\Pi_{p,r}(m) \equiv S_r(m) \pmod{p} \tag{34}$$

is legitimate coefficientwise after denominators are cleared. For primes dividing the chosen denominator, the raw congruence (31) *remains valid and no cancellation should be performed.*

Proof. For fixed r, the quantity $e_r(1, \ldots, m-1)$ is a polynomial in m of degree $2r$. One direct justification uses Newton's identities

$$re_r = \sum_{j=1}^{r} (-1)^{j-1} e_{r-j} \sum_{t=1}^{m-1} t^j,$$

together with Faulhaber's polynomial formulas for the power sums; induction on r gives polynomiality, and the leading term is $m^{2r}/(2^r r!)$. This polynomial vanishes at $m = 1, 2, \ldots, r$, because fewer than r nonzero slots are present. Hence it is divisible in $\mathbb{Q}[m]$ by $m - 1_{\underline{r}}$, proving that S_r is a polynomial of degree r.

By theorem 10.1, $\Pi_{p,r}$ is polynomial over $\mathbb{Q}[p, \boldsymbol{\lambda}]$. Specializing the recurrence at the formal value $p = 0$ removes every interaction term and sends ℓ_n to n, so the family becomes exactly the free product $\prod_{j=1}^{n}(1 + ju)$. Therefore $\Pi_{0,r} = S_r$. Polynomial division in the indeterminate p gives (33). Clearing rational denominators yields some integer D_r, and coefficientwise reduction modulo primes not dividing D_r gives (34). If $p \mid D_r$, only the pre-division raw theorem is automatic. □

Remark 12.3 (Exact information versus shadow information)**.** *The same parameter that is recovered exactly from one normalized row by Corollary 11.2 disappears from the entire raw mod-p jet by theorem 12.1. Thus the hierarchy exhibits a systematic exact-versus-shadow information loss at every order.*

13 The all-order closure theorem

We can now package the preceding results into one statement.

Theorem 13.1 (General higher-order chronological-carry theorem)**.** *Fix $k \geq 3$. Within the typed terminal-block class of Definition 4.1:*

(i) *at every order $r \leq k$, the chronological path has exactly one connected support type, namely the consecutive r-block;*

(ii) *after all lower sources are fixed, the sole new degree-r recurrence datum is the source sequence $a_{r,n}$;*

(iii) *the recurrence has the exact lower-Hessenberg realization* (11)–(13)*;*

(iv) *the r-th fixed column is insensitive to every subdiagonal deeper than $r-1$;*

(v) *normalized columns satisfy the boundary/transport system* (15)–(16)*;*

(vi) *every connected source sequence is reconstructed exactly by* (17)–(18)*;*

(vii) *in the stationary p-divisible subclass, the extension space through order k is $\mathbb{Z}^{k-3}$, with operator* (22)*;*

(viii) *stationary normalized columns are polynomials of exact degree equal to their column index;*

(ix) *every p-divisible member has the universal raw Stirling shadow* (30)*.*

No additional connected support type or deeper band datum is required to determine the k-jet.

Proof. Part (i) is Lemma 2.1. Parts (ii) and the absence of an additional order-r source are theorem 4.2. Part (iii) is theorem 5.2. Part (iv) is theorem 6.1. Parts (v) and (vi) are theorems 7.1 and 7.2. Part (vii) is theorems 8.2 and 9.1. Part (viii) is theorem 10.1. Part (ix) is theorem 12.1. Finally, a new connected support type cannot appear because connected subsets of the chronological path are already classified by Lemma 2.1, and a deeper subdiagonal cannot alter the k-jet by theorem 6.1. This closes the order induction. □

14 Low-order recovery and the fifth-order checkpoint

The general theorem recovers the audited low orders exactly and makes the next order transparent.

14.1 Order three

There are no free higher parameters. Equation (19) is exactly (1), and (22) is (6).

14.2 Order four

There is one free parameter λ_4. Equation (19) becomes (3); the new matrix band is

$$H_{i+3,i} = -\lambda_4 p\, i(i+1)(i+2),$$

and the operator is (7). The translation theorem gives

$$\Pi_{p,4}^{(\lambda_4)}(m) - \Pi_{p,4}^{(\mu_4)}(m) = \frac{p(\lambda_4 - \mu_4)}{4}.$$

This is precisely the fourth-order classification and identifiability mechanism.

14.3 Order five

At order five a second independent stationary parameter appears. The recurrence is

$$\begin{aligned} P_{p,n}^{[5]}(u) &= (1+\ell_n u)P_{p,n-1} + p(n-1)\ell_{n-1}u^2 P_{p,n-2} - pn - 1_{\underline{2}}u^3 P_{p,n-3} \\ &\quad + \lambda_4 pn - 1_{\underline{3}}u^4 P_{p,n-4} - \lambda_5 pn - 1_{\underline{4}}u^5 P_{p,n-5} \pmod{u^6}. \end{aligned} \tag{35}$$

The new fifth-order source occupies the fourth subdiagonal

$$H_{i+4,i} = -\lambda_5 p\, i(i+1)(i+2)(i+3)$$

and contributes the operator term $-\lambda_5 p X^4$. Changing only λ_5 translates the fifth normalized column by

$$-\frac{p(\lambda_5 - \mu_5)}{5}.$$

The sign alternation in the recurrence is therefore converted into a common negative sign in the stationary Hessenberg bands.

Table 1: The first orders of the stationary hierarchy. The numerical source is additional arithmetic data; only its support type is fixed by the path geometry.

order r	recurrence source a_r	Hessenberg band	normalized translation per unit λ_r
3	$-p$	$-p\, i^{\overline{2}}$	fixed baseline
4	$+p\lambda_4$	$-p\lambda_4\, i^{\overline{3}}$	$+p/4$
5	$-p\lambda_5$	$-p\lambda_5\, i^{\overline{4}}$	$-p/5$
6	$+p\lambda_6$	$-p\lambda_6\, i^{\overline{5}}$	$+p/6$

15 Symbolic regression checks

The proofs above are symbolic and do not rely on finite computation. We nevertheless used computer algebra as an independent regression check of signs, indices, and boundary conventions.

For a generic five-band matrix with symbolic p, λ_4, λ_5, direct determinants $\det(I_n + uH_n)$ were compared with recurrence (35) for $1 \le n \le 6$; every difference simplified identically to zero. The normalized transport equations were checked symbolically through column five. Finally, keeping λ_4 fixed and changing only λ_5 gave

$$\Pi_{p,5}^{(\lambda_5)}(m) - \Pi_{p,5}^{(\mu_5)}(m) = -\frac{p(\lambda_5 - \mu_5)}{5}$$

for successive admissible rows, matching theorem 11.1. These checks are consistency tests only.

16 Related work and novelty boundary

Several classical theories neighbor the present construction, and separating them from the family-specific claims is important.

Carries. Classical carry processes have rich matrix structure; Holte's "amazing matrix" is a standard example [8], and connections with shuffling and symmetric functions were developed by Diaconis and Fulman [5]. The present chronological carry family is not a new version of the classical base-b carry Markov chain. Its defining data are the ℓ_n-weighted terminal-block recurrence and the connected arithmetic sources on an ordered correction path.

Hessenberg production matrices and lattice paths. Banded lower-Hessenberg matrices and higher-order recurrences are classical. Coussement and Van Assche use banded Hessenberg recurrence matrices in multiple orthogonality [4]. Flajolet's path interpretation of continued fractions is foundational [6]; more recent work develops branched continued fractions and height-dependent m-Dyck paths [9]. Sokal emphasizes the role of lower-Hessenberg, not necessarily tridiagonal, production matrices in multiple and d-orthogonality [11]. Our determinant expansion is elementary in this broader setting. The family-specific content is the exact source-to-subdiagonal dictionary (13), the inverse normalized transport, and the stationary parameter geometry.

Stirling numbers. Unsigned Stirling numbers of the first kind and their elementary-symmetric realizations are classical; see, for example, Graham, Knuth and Patashnik [7]. Equation (31) is not claimed as a new Stirling identity. The point here is that all exact connected parameters of the p-divisible hierarchy collapse to that same shadow.

Connectedness and cumulant language. Cumulants are classically organized by Möbius inversion on the partition lattice [10, 12]. This motivates the informal word "connected", but none of the proofs above identifies CCR weights with probabilistic cumulants. Terminal-block separation is an internal deterministic recurrence statement on an ordered path.

Position of the contribution. The contribution is therefore not a new general carry, cumulant, or Hessenberg formalism. It is the all-order theorem package that we call the *CCR-k General Theory* for the specific chronological terminal-block family: one typed connected source per order, exact higher-band realization, a coefficientwise light cone, forward and inverse normalized transport, polynomial fixed columns, exact parameter translation and identifiability, and a universal shadow that forgets all p-divisible connected parameters. The lattice $\mathbb{Z}^{k-3}$ is the natural coordinate description of the stated stationary subclass; the stronger structural content is that the source coordinates are dynamically visible and reconstructible in the exact theory while becoming invisible in the common mod-p shadow. A targeted literature search across classical carry processes, lower-Hessenberg and production-matrix recurrences, and Stirling-number structures located adjacent theories but no direct prior result establishing this complete all-order source–band–transport–shadow classification. We therefore restrict the novelty claim to this family-specific theorem package and make no absolute priority claim for its classical ingredients.

17 Synthesis

The CCR-k General Theory has a simple structural message. Chronological path geometry is rigid enough to determine where a new connected interval can live, but not strong enough to determine its numerical weight. Once that weight is admitted as typed arithmetic data, the entire hierarchy becomes exact and recursively observable.

At every order r, four descriptions of the same new datum coexist:

$$\boxed{\begin{array}{c} \text{connected interval source } a_{r,n} \\ \Updownarrow \\ \text{terminal recurrence block } n-1_{\underline{r-1}}a_{r,n}u^rP_{n-r} \\ \Updownarrow \\ \text{Hessenberg subdiagonal } (-1)^{r-1}i^{\overline{r-1}}a_{r,i+r-1} \\ \Updownarrow \\ \text{source term in normalized-column transport.} \end{array}}$$

In the stationary p-divisible class, this reduces to one integer λ_r per order. The exact normalized column sees λ_r immediately through the constant translation $(-1)^r p\lambda_r/r$, while reduction modulo p sees none of it. The free Stirling shadow is therefore universal precisely because it is an information-losing projection.

The resulting all-order spine is

$$\boxed{\begin{array}{c}\text{chronological locality + typed connected arithmetic data}\\ \Longrightarrow\ \text{higher-band Hessenberg transport}\\ \Longrightarrow\ \text{exact identifiability + universal Stirling shadow.}\end{array}}$$

A Proof-dependency ledger

For audit purposes, the logical dependencies of the main results are summarized in table 2.

Table 2: Dependency ledger for the all-order theorem package.

Result	Direct inputs	Role
Connected interval support	path graph	one support type per order
Terminal-block separation	Definition 4.1	isolates the new order-r source
Hessenberg realization	source normalization + last-row determinant expansion	recurrence/band equivalence
Light cone	lower-Hessenberg displacement bound	excludes deeper-band contamination
Normalized transport	recurrence coefficient extraction + falling-factorial cancellation	row evolution of fixed columns
Source reconstruction	normalized transport	inverse identifiability
Stationary classification	one source type/order + stationarity + p-divisibility	parameter lattice $\mathbb{Z}^{k-3}$
Operator realization	divided-power basis identities	compact all-band representation
Polynomiality	stationary transport + invertibility of $\mathcal{T}_j$	exact degree and leading coefficient
Translation law	subtraction of two stationary transports	one-row parameter identifiability
Stirling shadow	p-divisibility + $\ell_n \equiv n \pmod p$	universal information-losing projection
Main theorem	all preceding rows	all-order closure

B Scope safeguards and nonclaims

The following statements are part of the mathematical scope of the paper.

1. The original pushdown correction process supplies a motivating chronology and path-local support geometry. The numerical CCR sources are an added arithmetic polarization.
2. The typed terminal-block factorization (9) is an explicit model class. We do not assert that every possible path-local deformation must factor in this form.
3. The stationary p-divisible classification is a classification *inside that stated class.*

4. The primitive same-sign point $\lambda_r = 1$ is a normalization convention, not a theorem of machine-forced uniqueness.
5. The universal raw mod-p shadow is stronger and safer than any normalized congruence. Normalized polynomial reduction requires denominator control as in Proposition 12.2.
6. Classical results on cumulants, Möbius inversion, carry matrices, Stirling numbers, continued fractions, and lower-Hessenberg recurrences are used only as context or standard tools.

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Ethics Approval and Consent to Participate. Not applicable.

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